

CSC 648/848 Software Engineering Spring 2026

Pathwise

Section 2

Team 2

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Milestone 1

Checklist

- ☒ Executive Summary
- ☒ Personas and main Use Cases
- ☒ Main Data Items and Entities:
- ☒ Initial list of functional requirements
- ☒ List of non-functional requirements
- ☒ Competitive analysis
- ☒ High-level system architecture and technologies used

History Table (Revision)

Name	Revision Summary	Revision Date
Laura	Started on <i>Executive Summary</i>	2/10/2026
Omeid	Completed <i>Initial List of Functional Requirements</i> and copied <i>Non-Functional Requirements</i>	2/10/2026
Damian	Started on <i>Main Data Items and Entities</i>	2/12/2026
Cielina	Started on <i>Personas and Use Cases</i>	2/12/2026
Brandon Sanchez	Started <i>High-level System Architecture and Technologies Used</i>	2/13/2026
Jason Le	Started <i>Competitive analysis</i>	2/14/2026
Damian	Finished <i>Main Data Items and Entities</i> ; updated document formatting	2/17/2026
Laura	Finished <i>Executive Summary</i>	2/17/2026
Brandon Sanchez	Finished <i>High-level System Architecture and Technologies Used</i>	2/17/2026
Mikias	Revised <i>Competitive Analysis</i>	2/17/2026

Executive Summary

Our team will be working on Pathwise, a website that hosts a variety of features designed to help students succeed. This project is designed to help make life easier for students with the help of AI. This platform will host services that help guide users learning to use AI on how to ethically use it, how powerful of a tool it is and more. The website will mainly focus on providing tools to students for success.

We will be building a platform accessible through the web that allows for users, mainly students, to be able to access tools that can help make their lives easier. The platform starts off at the interactive dashboard where the user is able to manage goals at a glance. It will include goals, projects, learning activities and personal development. The main feature that we would like to highlight is the built in search engine which allows for an easy way to search for project templates, academic resources, opportunities on campus, etc. In addition to a search function, there will be keywords to filter results. To make it a unique experience for users, users will be able to add their interests, challenges, etc. Goals and progress will be shown on interactive dashboards that provide a visual on progress made. Overall, the platform is designed to help students with growth in their personal and professional life as well as how to use AI in pursuit of their goals.

It will be built by students for students. The team is made up of Omeid, our lead, Cielina, Laura, Brandon, Damian, Jason and Mikias. As students ourselves, we understand how our peers may feel overwhelmed by school. Our goal is to create a platform that can do good for other students by providing the tools needed for success. We are passionate about creating something that positively impacts others.

Personas and Main Use Cases

Persona 1: SFSU Undergraduate Student

The SFSU undergraduate student is a full-time student, balancing both academic and commitments outside of school. The student experiences stress from managing time in between school and work. Ensuring that they are meeting deadlines, studying, and fulfilling their responsibilities. Their goal is to stay organized with classwork, time to review, and submitting applications for internships. They feel overwhelmed with the multiple platforms and struggle to meet their needs and fulfill tasks. They want to be able to manage their time and tasks better.

Persona 2: SFSU Graduate Student

The SFSU Graduate student is enrolled in the master's program. They manage advanced coursework, research projects, and extracurriculars. They are also employed part time and have commitments outside of school. They are managing their time for their responsibilities for school and outside of that. Their goals are to graduate, maintain a proper schedule, make time to study, and work. They are comfortable with research tools and using the computer. The SFSU Graduate's struggle is organizing their time, research, and tasks properly. They want to have a system that will allow them to access their schedule with ease.

Persona 3: SFSU International Student

The SFSU International student may be either in an undergraduate or graduate program. The challenges they face are visa compliance and cultural adaptation. The student manages their tasks and time around studying, applying for internships, and extracurriculars. Their goal is to pass their classes, maintain visa status, graduate and secure internships or job opportunities. The student uses web based tools to guide them around campus and understand coursework. They use it as a way to interpret and communicate with others. They often experience stress and anxiety about making mistakes that would impact their student status. They want to be able to have access to information easily that they can understand and use efficiently.

Persona 4: SFSU Faculty Member

The SFSU Faculty Member is responsible for teaching, advising, and supporting students for academic success. They manage multiple classes, responsibilities, and tasks to fulfill the coursework objectives. The member's goal is to help students succeed while encouraging learning habits and applicable skills for the student's careers outside of school. They are familiar with university systems that allow them to manage documents, grades, and communication tools. They are comfortable with using web browsers, applications, and various technology platforms. The faculty members are concerned that students lack awareness of resources or fail to plan effectively outside of class. They would like to find a structured guide for students to have access to in order to achieve academic success.

Persona 5: SFSU University Support Staff

SFSU University Support Staff work in offices that are responsible for academic advising, career services, tutoring, and student counseling. They interact with a diverse population of students and focus on supporting the student's needs to succeed in class and graduate. Their goal is to increase engagement for students to access campus services and provide guidance for students who need assistance. They are comfortable using web based systems and databases. They are not technical experts, they like to access platforms that are easy to read and use. The staff's recurring problem is that students seek assistance too late and ask repetitive questions. They would like a tool that would announce or alert students regarding assistance and specific dates. They need a system that will support student engagement and foster assistance provided by campus.

Main Use Cases

UC_1: Establishing and Organizing Goals for the Semester

At the beginning of a semester, an undergraduate or graduate student feels uncertain about how to balance coursework, personal responsibilities, and career preparation. The student uses the application to clarify academic, personal, and professional goals and organizes them into a structured plan. Throughout the semester, they revisit these goals to adjust priorities as deadlines shift or new responsibilities arise. This ongoing organization helps the student maintain direction and reduce stress by connecting daily tasks to larger objectives.

UC_2: Exploring and Selecting Relevant Resources

When a student wants to improve study strategies, prepare for internships, or understand graduation requirements, they use the application to explore relevant campus and academic resources. Instead of searching across multiple disconnected websites, they review options aligned with their current goals and evaluate which opportunities best fit their needs. The student selects appropriate resources and incorporates them into their broader academic or professional plan.

UC_3: Reviewing Progress and Reflecting on Performance

Midway through the semester, a student wants to understand whether they are on track. They review their goals and responsibilities within the application and reflect on completed milestones and upcoming deadlines. By seeing their progress in one place, they identify areas that require attention and adjust priorities accordingly. This reflective process encourages proactive decision-making rather than last-minute crisis management.

UC_4: Supporting Advising and Academic Guidance

During an advising session, a faculty member or support staff member engages a student in a structured conversation about goals and progress. The application serves as a shared reference point for discussing academic plans, career preparation, and available support services. This

interaction helps students take ownership of their planning while allowing faculty or staff to provide targeted guidance without needing to manually track each student's independent activities.

UC_5: Managing Academic and Compliance Responsibilities for International Students

An international student uses the application to maintain awareness of both academic milestones and visa-related responsibilities. By organizing these interconnected obligations together, the student gains a clearer understanding of how coursework, employment opportunities, and institutional requirements relate to long-term success. This structured overview reduces anxiety and helps the student feel more confident navigating academic and professional pathways.

Main Data Items and Entities

This section defines the key data entities and terms used throughout the Pathwise application. These terms and their names must be used consistently from this point forward in all documents, user interface, software components, and database elements.

User Types and Privileges

Registered User: A student or individual who has created an account in the Pathwise system. Registered users can create and manage goals, organize projects, save resources, share content with the community, send in-site messages, and access all personalized features including the dashboard and search history. They have full control over their personal data.

Guest User: A visitor who has not created an account. Guest users can browse publicly available content and use the search engine to discover resources and templates, but cannot save content, create goals or projects, or access personalized features.

Faculty/Staff User: A registered user with additional privileges to publish official academic resources, tools, and workflows. Faculty and university support staff can categorize and tag institutional content for student discovery.

Core Data Entities

Goal: A user-defined objective that the student wants to achieve. Each goal has a title, description, category (academic, career, health, personal development, etc.), target completion date, and status (active, completed, paused). Goals serve as the primary organizing structure for a user's work in Pathwise.

Project: A structured plan with milestones designed to help achieve a specific goal. Each project has a name, description, associated goal, timeline, and completion status. Projects break down larger goals into manageable phases.

Milestone: A specific checkpoint or phase within a project. Each milestone has a title, description, target date, and completion status. Milestones enable progress tracking and visualization.

Resource: External or internal content that supports goals or projects, such as articles, videos, tools, guides, or tutorials. Each resource has a title, URL or location, description, category, tags, an AI-enabled flag indicating whether the tool uses artificial intelligence, and can be linked to specific goals or projects. Resources are discoverable through the search engine and can be bookmarked by users.

Template: A pre-built goal or project framework shared by users or published by faculty/staff. Examples include "Freshman Academic Success Plan" or "Job Search Strategy." Each template has a title, description, category, creator, and can be copied and customized by other users.

Workflow: A structured process or step-by-step guide for accomplishing specific tasks. Workflows can be attached to goals or projects and shared in the community. Each workflow has a title, description, steps, category, and tags.

Reflection: A user journal entry to record thoughts, insights, or progress updates. Each reflection has a date, text content, optional goal or project association, and can help users track their development over time.

Message: A private communication sent between two registered users within the Pathwise platform. Each message has a sender, recipient, text content, timestamp, and read status. Messages support direct communication between students and faculty or support staff without relying on external email clients.

ActivityLog: A system-generated record of a user's interactions and progress over time, including goals created, milestones completed, resources saved, and reflections submitted. Each log entry has a user reference, action type, associated entity, and timestamp. ActivityLogs are used to generate the analytics and summary reports displayed to users over a semester.

Dashboard: A personalized view displaying the user's active goals, current projects, milestone progress, saved resources, and activity summary. The dashboard provides an at-a-glance snapshot of the user's current priorities and progress.

Category/Tag: Metadata used to organize and filter content. Categories represent broad areas (academic, career, health, personal development) while tags are specific descriptors (time management, resume writing, mental health). Tags enable powerful search filtering.

Data Relationships

- Goals contain one or more Projects
- Projects contain multiple Milestones
- Resources, Templates, and Workflows can be linked to Goals or Projects
- Reflections can reference specific Goals or Projects
- Categories and Tags classify Resources, Templates, Workflows, and Goals
- Messages are exchanged between Registered Users and are not associated with Goals or Projects
- ActivityLogs are system-generated per Registered User and aggregate actions across Goals, Projects, Milestones, Resources, and Reflections
- All content is owned by and private to the Registered User who created it, unless shared publicly

Terminology Usage

These terms must be used consistently throughout the application. We will always use "Goals" (not "Objectives"), "Resources" (not "Links"), "Templates" (not "Blueprints"), "Milestones" (not "Checkpoints"), "Reflections" (not "Journal Entries"), "Workflows" (not "Processes"), and "Messages" (not "Emails" or "Chats"). This ensures clear communication in the UI and documentation.

Initial List of Functional Requirements

FR_01: The system shall allow users to create and manage a personal account with authenticated login.

Relevant Personas: Undergraduate Student, Graduate Student, International Student, Faculty Member, University Support Staff

Use Case Example: A student registers for an account to access personalized dashboards and goal tracking features for their thesis project.

FR_02: The system shall allow registered users to create, edit, and delete short-term and long-term goals.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: A senior undergraduate student creates a short-term goal to apply and secure an internship.

FR_03: The system shall allow users to organize goals into structured projects and milestones.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: A student organizes their project into milestones such as proposal approval, data collection, and implementation.

FR_04: The system shall allow users to attach discovered resources, tools, or workflows to specific goals or projects.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: An undergraduate student attaches interview preparation guides and coding practice tools to their internship project.

FR_05: The system shall provide a searchable discovery engine that allows users to search for goals, resources, workflows, tools, events, and templates.

Relevant Personas: All Registered Users, General Public User

Use Case Example: A public user searches for “study strategies for finals” and browses categorized templates before deciding to register.

FR_06: The system shall allow users to filter search results by category, use case, skill area, content type, AI-enabled status, and cost type.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: An undergraduate student filters results to show career-related, AI-enabled, technical skill resources.

FR_07: The system shall display search results with relevant metadata such as tags, ratings, and descriptions.

Relevant Personas: All Users

Use Case Example: A graduate student reviews ratings and tags before selecting a research workflow template.

FR_08: The system shall allow users to bookmark or save discovered resources for later access.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: An international student bookmarks visa information resources and career center links for future reference.

FR_09: The system shall provide a personalized dashboard displaying user goals, progress, milestones, and saved resources.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: An undergraduate logs in and views their dashboard to see progress toward completing internship applications.

FR_10: The system shall visually display goal progress and milestone completion status.

Relevant Personas: Undergraduate Student, Graduate Student

Use Case Example: A graduate student reviews a visual timeline of thesis milestones to monitor completion progress.

FR_11: The system shall allow users to complete structured reflections on academic, professional, or personal development.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: An undergraduate student completes a weekly reflection about workload and stress levels during midterms.

FR_12: The system shall track and summarize user contribution history over time.

Relevant Personas: Undergraduate Student, Graduate Student

Use Case Example: A graduate student reviews past reflections to evaluate productivity patterns during research intensive months.

FR_13: The system shall allow users to enter personal interests, challenges, workload levels, and aspirations to support personalization.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: A student enters interests in software engineering and notes challenges with networking for internships and jobs.

FR_14: The system shall provide personalized recommendations for resources, workflows, and goal templates based on user profile and activity.

Relevant Personas: Undergraduate Student, Graduate Student, International Student

Use Case Example: Based on activity, the system recommends fellowship opportunities and research tools to a graduate student.

FR_15: The system shall allow users to browse and explore public community-shared goal templates and workflows.

Relevant Personas: Undergraduate Student, Graduate Student, International Student, General Public User

Use Case Example: A first-year student browses a “Freshman Academic Success Plan” template shared by peers.

FR_16: The system shall allow users to share resources, tools, or goal templates with other users.

Relevant Personas: Undergraduate Student, Graduate Student

Use Case Example: A graduate student shares a structured research planning workflow with other students in their department.

FR_17: The system shall provide in-site messaging functionality to enable communication between registered users.

Relevant Personas: All registered users

Use Case Example: A student messages a career center advisor regarding resume improvements.

FR_18: The system shall allow users to rate and tag shared content within the platform.

Relevant Personas: Undergraduate Student, Graduate Student

Use Case Example: An undergraduate rates a study template and tags it as “time management” to assist other users.

FR_19: The system shall allow faculty or university support staff to publish categorized academic and career resources.

Relevant Personas: Faculty Member, University Support Staff

Use Case Example: A career center staff member publishes categorized internship and workshop resources for student access.

FR_20: The system shall provide analytics or summary reports reflecting user progress, activity trends, and goal development.

Relevant Personas: Undergraduate Student, Graduate Student

Use Case Example: At semester end, a graduate student views a summary report highlighting completed milestones and research progress.

FR_21: The system shall allow general public non-registered users to browse publicly available categorized content.

Relevant Personas: General Public User

Use Case Example: A prospective student browses public career preparation resources without creating an account.

FR_22: The system shall display ethical labeling or indicators distinguishing AI-enabled tools from non-AI resources.

Relevant Personas: Undergraduate Student, Graduate Student, International Student, Faculty Member

Use Case Example: A faculty member reviews AI-labeled tools before recommending them to students for responsible usage.

Non-Functional Requirements

(Copied Exactly from Team Project High Level Description.pdf)

1. Application shall be developed, tested, and deployed using tools and servers approved by Class CTO and as agreed in M0.
2. Application shall be optimized for standard desktop/laptop browsers and must render correctly on the two latest versions of at least two major browsers.
3. Selected application functions must render well on mobile devices (this is a plus).
4. Data shall be stored in the team's chosen database technology on the team's deployment server.
5. Privacy of users shall be protected, and all privacy policies shall be clearly communicated to users.
6. The language used shall be English.
7. Application shall be very easy to use and intuitive.
8. Google Maps and Google Analytics shall be added where appropriate (e.g., campus resources, study locations, event discovery, usage analytics).
9. No external e-mail clients shall be allowed. You shall use in-site webmail only.
10. Pay functionality, if any, shall not be implemented nor simulated in the UI.
11. Site security: basic best practices shall be applied (as covered in the class).
12. Modern Software Engineering processes and practices shall be used as specified in the class, including collaborative and continuous software development.
13. The website shall prominently display the following exact text on all pages at the top of the WWW page: "SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only"
14. AI-powered assistant and intelligent search enhancements are optional (bonus requirements) and must be implemented modularly.

Competitive Analysis

Features	Notion (Student context use)	MyStudyLife	Handshake	Navigate (EAB)	Pathwise (Our App)
Academic Tasks Management	Yes	Yes	No	Yes	Yes
Goal-Project- Milestone Framework	Manual	Limited	No	Limited	Yes
Aggregation of Campus Resources	No	No	Yes	Limited	Yes
Faculty/Staff Advising	No	No	Limited	Yes	Yes
Career Opportunities Discovery	No	No	Yes	Limited	Yes
All-In-One	No	No	No	No	Yes

The competitors mentioned such as Handshake, Notion (from a student use-case perspective), MyStudyLife, and Navigate (EAB) only focus on a subset of the total university student experience and all the features are fragmented across these platforms. They are generally catered toward a specific goal, with Notion serving as reconfigurable note taking place, Handshake for student career opportunities, Navigate (EAB) for student advising by staff/faculty, and MyStudyLife for tasks progress tracking. As you can see, we can take each of their strongest features and combine it into a one unified system with the addition of an integrated campus discovery system and as well as a refined Goal-Project-Milestone framework feature. So with a clear intent, Pathwise will be designed as the one solution with an integrated academic planning with task management, Goal-Project-Milestone framework (serves as a folder for tasks) and student/staff advising portal, as well as career opportunities and campus resource aggregation with all coming to serve as a unified holistic system.

High-level System Architecture and Technologies Used

Main SW Components:

- Backend Framework: Express.js (Node.js)
- Frontend: HTML, CSS, JavaScript
- Database: MySQL 8.0.45
- Server: Nginx 1.24.0 (Ubuntu)
- Version Control: Git/GitHub
- Authentication: AWS IAM for team member access
- User authentication: Session Based ID
- Package Manager: NPM

Supported Browsers

- Chrome
- Firefox
- Safari

Deployment Platform

- Operating System: Ubuntu 24.04 LTS
- Web Server: Nginx 1.24.0
- Server Hosting: AWS EC2
- Database Hosting: AWS RDS (MySQL 8.0)
- Runtime: Node.js 18.19.1

Team and Roles

Role	Team Member
Front End Lead	Damian Perez
Back End Lead	Jason Le
Database Engineer	Omeid Nadery
GitHub Master	Laura Wong